

SMART INDIA HACKATHON 2026 — OFFICIAL 5-MINUTE PRESENTATION

PROJECT A.U.R.A. | TEAM DELTA | Panimalar Engineering College (Autonomous)

Panel Head: Dr. S. Balaji (Professor, Dept of CSE)	Team Name: TEAM DELTA
Schedule: 7.9.2026, 6:00 PM – 8:00 PM	Problem Statement ID: SIH1566
Meeting Link: meet.google.com/gad-jeft-gtj	Designated Speakers: Kingston J & Nithyashree V

VERIFIED SENSOR RANGES & PENETRATION DEPTHS

Sensor Modality	Hardware Model	Tested Depth / Range	Medium Penetrated	Target Detected
Microwave Radar	RCWL-0516 (3.18 GHz)	0.5m – 2.5m rubble (Up to 6m open air)	Concrete rubble, brick, drywall, wood	Chest breathing flutter & body motion
Seismic Transducer	Piezoelectric Disc (Pin 35)	3.0m – 8.0m along steel/pipes (1m – 2.5m soil)	Solid concrete slabs, pillars, rebar	Rhythmic survivor tapping with stones/metal
Acoustic Array	MAX9814 Electret (Pin 32)	2.0m – 5.0m reach (along void air fissures)	Air pockets, cracks, vent cavities	Faint whispers, crying, and vocal calls
Bio-Scent VOC	MQ-135 Gas Array (Pin 34)	1.0m – 2.5m radius (diffusion in voids)	Enclosed air pockets, collapse voids	Exhaled breath VOCs, CO2 & ammonia
Structural IMU	MPU-6050 6-Axis (I2C)	0m – 1.5m chassis ground	Rubble ground contact plane	Debris micro-shifting & aftershock safety

SLIDE 1: INTRODUCTION & TEAM CREDENTIALS

■ [0:00 – 0:45] SPEAKER: KINGSTON J

(Camera ON. Confident, warm, formal greeting. Slide 1 visible on screen)

"Good evening respected Panel Head Dr. Balaji Sir and honorable judges.

We are Team DELTA from Panimalar Engineering College (Autonomous).

We are presenting for Problem Statement ID: SIH1566, under the theme Disaster Management and Robotics, categorized as a Hardware and Software Integrated System.

Our Problem Statement Title is: 'Rapid Non-Invasive Subterranean Victim Localization & Void Life Detection in Post-Disaster Structural Collapse Scenarios.'

My name is Kingston J, and presenting with me today is my teammate Nithyashree V. Together with our teammates Jaffer Rilwaan, Hannah Blessy, Kathiravan, and Giridhar, we have developed Project A.U.R.A. — the Autonomous Universal Rescue & Analysis Node."

SLIDE 2: THE DISASTER GROUND REALITY & THE 72-HOUR WINDOW

■ [0:45 – 1:45] SPEAKER: KINGSTON J

(Tone shifts to serious, compelling narrative about real collapse rescue operations)

"Let me take you directly to ground zero of an earthquake or a landslide disaster, like Wayanad or Turkey.

When multi-story concrete structures collapse, human beings get trapped in small void spaces beneath tons of debris. Trapped survivors face the critical 72-Hour Golden Window. After three days, lack of water, dust inhalation, and toxic air cause survival rates to drop drastically.

When our emergency NDRF and SDRF teams arrive on the scene, they face a complete sensory blackout. Flying drones with thermal cameras are blind because thermal infrared cannot see through concrete slabs. Mobile phones and radios are dead because dense steel rebar blocks wireless signals.

Heavy foreign military equipment weighs **15 to 20 kilograms**, costs an enormous **15 Lakh Rupees**, and can only penetrate **3 to 4 meters** in simple brick walls.

Because rescue teams have no sub-surface vision, excavators are forced to dig blindly. That blind digging shakes the rubble, collapses the fragile air pockets, and accidentally crushes the very survivors teams are trying to save.

We asked ourselves: *Why should human lives be lost just because existing tools cannot penetrate concrete?* That is the problem Team DELTA set out to solve with Project A.U.R.A."

SLIDE 3: HOW PROJECT A.U.R.A. WORKS & OUR SENSORS

■ [1:45 – 3:00] SPEAKER: NITHYASHREE V

(Nithyashree speaks clearly and with energy, holding up the hardware probe for the panel)

"Thank you, Kingston. Respected judges, think of Project A.U.R.A. as a doctor's stethoscope, an acoustic listening ear, a penetrative radar eye, and a trained search dog — all combined into one lightweight, handheld node.

Instead of guessing, A.U.R.A. combines four distinct physical senses driven by an ESP32 dual-core processor running FreeRTOS:

1. Our 3.18 GHz Microwave Doppler Radar on Pin 33: While camera light bounces off rubble, radio microwaves travel directly through drywall, brick, and concrete up to **2.5 meters deep** — and up to **6.0 meters in open void spaces**. It detects the unconscious rise-and-fall chest breathing and body twitching of a trapped survivor.

2. Our Piezoelectric Seismic Geophone on Pin 35: It presses directly against concrete slabs. When a trapped victim knocks with a stone or pipe, it captures those shockwaves traveling through steel pipes and rebar up to **8.0 meters away**.

3. Our Low-Noise Acoustic Microphone on Pin 32: It listens along rubble cracks and air fissures between **2.0 to 5.0 meters**, sampling sound 500 times a second to detect faint whispers, gasps, and crying.

4. Our MQ-135 Bio-Scent Gas Sensor on Pin 34: It functions like a digital search dog, detecting exhaled carbon dioxide, ammonia, and human sweat VOCs within a **1.0 to 2.5 meter diffusion radius**, while warning rescuers if dangerous gas is leaking.

Finally, an MPU-6050 sensor monitors debris shifting to protect responders from secondary collapses, and an onboard 8-slide OLED screen lets a firefighter read victim depth directly on the hardware even without a phone or laptop."

SLIDE 4: DEPTH CALCULATION & OPERATIONAL TRIAGE LEVELS

■ [3:00 – 4:00] SPEAKER: KINGSTON J

(Kingston explains the math simply and highlights the practical operational depth levels)

"Now, how does A.U.R.A. calculate where to dig?

It uses a well-established law of physics called energy attenuation: the further sound and shockwaves travel through concrete, the weaker they become. By combining the acoustic energy and seismic impulse peaks, A.U.R.A. computes victim distance using the formula:

Distance = 12 divided by square root of maximum energy, multiplied by 0.8 for true depth.

To make this simple and actionable for frontline firefighters, A.U.R.A. translates this into **Three Clear Operational Levels** on the screen:

- **Surface Level (Under 0.5 meters):** The survivor is right near the surface. Responders can dig immediately using manual hand tools like shovels and crowbars.
- **Mid-Debris Core (0.5 to 2.5 meters):** The victim is buried beneath reinforced structural slabs. Teams immediately deploy hydraulic spreaders, jacks, and concrete slab cutters.
- **Deep Void Cavity (2.5 to 6.0 meters):** The victim is trapped deep down in a basement void or foundation layer. Responders must shore up the walls first and drill a narrow oxygen pipe.

Every single measurement is genuine physical sensor data. Our ESP32 streams this over WiFi directly into **Google Gemini Flash AI**, which gives the commander a clean directive: *'Survivor located 2.4 meters deep at 320 degrees North-Northwest.'*

SLIDE 5 & 6: COST ECONOMICS, HONEST ENGINEERING & CONCLUSION

■ [4:00 – 4:45] SPEAKER: NITHYASHREE V

(Nithyashree shares the prototype demonstration, cost comparison, and honest engineering)

"As you can see on our live screen right now, our working prototype is online and streaming live telemetry over WiFi to our React 19 Tactical C2 Dashboard.

Let us examine Economics and Feasibility:

An imported foreign search system like Delsar costs **12 to 15 Lakh Rupees**. Because of that price tag, only elite central units have them. District and local fire stations have zero.

Project A.U.R.A. is manufactured at **under ₹3,500 Rupees — less than 42 US Dollars**. That means for the price of ONE imported machine, our government can equip **400 fire stations and rescue squads** across every district in India.

We also believe in honest engineering: **What are our physical limits?**

A.U.R.A. easily penetrates concrete slabs 20 to 30 cm thick, brick walls, and dry rubble. However, it cannot penetrate solid sheet-metal barriers or flooded waterlogged mud. But because we combine radar, tapping, sound, and scent, when one sensor faces a physical barrier, the other three take over.

Our Phase 2 is already tested: adding LoRa mesh swarm networking so multiple A.U.R.A. nodes communicate across an entire disaster field simultaneously."

GRAND FINALE CLOSING

■ [4:45 – 5:00] BOTH TOGETHER

KINGSTON: "Every second beneath the rubble is a battle between life and death. Project A.U.R.A. was created to ensure that no call for help ever goes unheard in the dark."

NITHYASHREE: "Representing Panimalar Engineering College, we are **Team DELTA**, and our working prototype is live and ready for your test. Thank you Dr. Balaji Sir and honorable judges, and we welcome your questions!"

3-MINUTE JUDGES' Q&A; PREPARATION GUIDE

Q1: How do you prove your sensor readings are real and not simulated?

Kingston: 'Sir, our prototype is streaming live right now. When I tap the table near our geophone, the impulse spikes on the screen, the tap counter increments, and decibels rise. In a quiet room, the acoustic spectrum reads SILENCE and victim count is zero. Every reading comes directly from ESP32 ADC1 registers (Pins 32, 34, 35) with zero synthetic floors.'

Q2: Why does your microwave radar penetrate concrete when optical drones fail?

Nithyashree: 'Optical cameras use visible light with nanometer wavelengths that get absorbed by concrete surfaces. Our RCWL-0516 uses 3.18 GHz microwave radio waves. These long radio waves travel right through non-metallic dielectric materials like concrete, brick, and wood, reflecting off moving human chest muscle.'

Q3: What happens if there is no internet or cellular connectivity at the disaster site?

Kingston: 'A.U.R.A. is 100% offline-first. The ESP32 hosts its own local Wi-Fi hotspot and web server. Responders connect directly without internet. Furthermore, the onboard 8-slide OLED screen lets a firefighter read depth, tap count, and gas levels directly on the probe even if all tablets fail.'

Q4: How did you manage to keep the cost under Rs. 3,500?

Nithyashree: 'Foreign systems use proprietary military FPGA chips. We replaced expensive hardware with smart software — running our 500Hz DSP on Core 1 of an ESP32-WROOM-32. Using commercial off-the-shelf sensors keeps our total component bill to exactly Rs. 3,420 INR.'